CONFIGURING THE SI5338 EVALUATION BOARD IN ZERO DELAY MODE

1. Introduction

The Si5338 can be configured in zero delay mode to help minimize the input-to-output delay. This application note provides details on setting up an Si5338 evaluation board (Si5338-EVB) in zero delay mode. In this example, the Si5338 will be configured to generate a 10 MHz output on CLK0A/B while locked to a 10 MHz input on IN1/2.

Figure 1 shows a block diagram of the Si5338 configured as described in the example above but with internal feedback (non-zero delay mode). This results in an input-to-output delay of approximately 2.5 ns (typ). Most of this delay is caused by the circuitry outside of the internal feedback path such as the MultiSynth dividers, output dividers, output drivers, input receiver, input divider, etc.

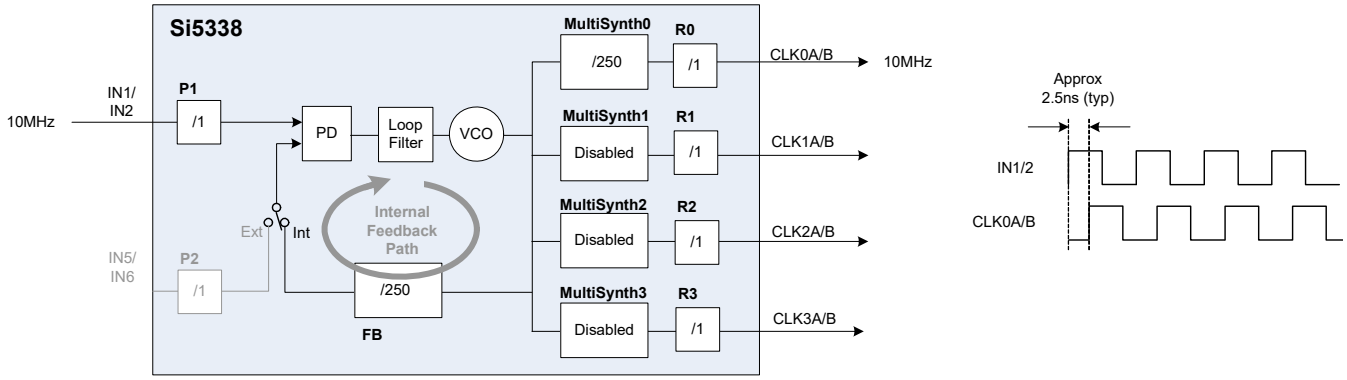


Figure 1. Si5338 Configuration In Standard PLL Mode

Configuring the Si5338 in zero delay buffer mode as shown in Figure 2 extends the feedback path across much of the circuitry that causes the input-to-output delay. This helps in minimizing the input-to-output delay. In this case, the feedback divider (FB) is replaced with one of the output MultiSynth dividers (M3), and the loop is closed externally by connecting one of the output clocks (CLK3A/B) to the feedback input (IN5/6) of the Si5338. Note that IN3 can be used when routing a single-ended feedback path. It is important to note that the divider value of the M3 divider (/250 in this example) must be equal to the divider value of the FB divider (/250 in this example) to properly close the loop. In other words, the objective is to replace the feedback divider with the output divider.

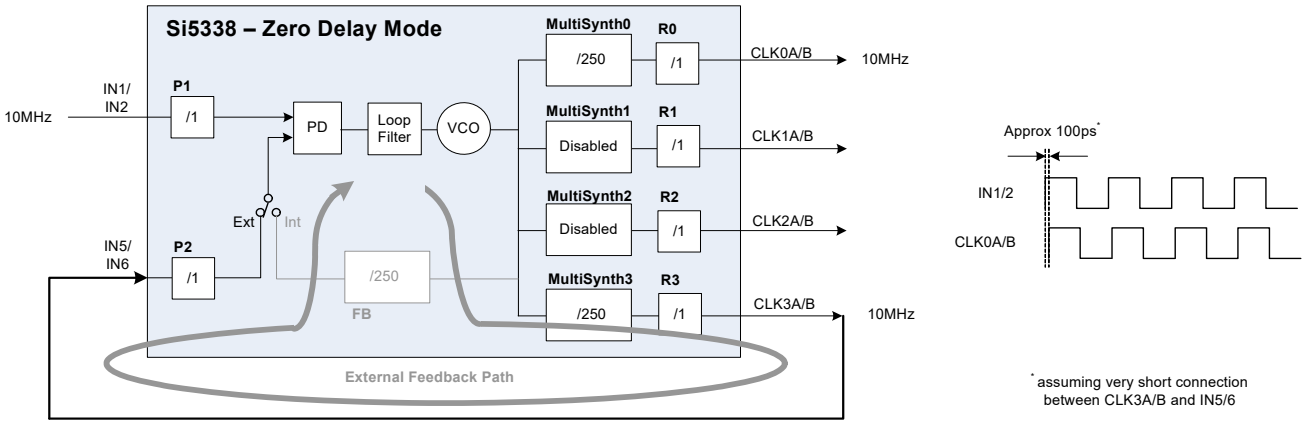


Figure 2. Si5338 Configured in Zero Delay Mode

2. Setup

The first step in configuring the Si5338-EVB for zero delay mode is making a connection between the CLK3A/B output and the IN5/6 input as shown in Figure 2. Using short and good quality SMA cables will achieve the best results.
Using the Si5338 MultiSynth Clock Programmer Software, configure the Frequency Plan Tab, the Output Driver Tab, and the Advanced Tab as shown in the following sections.

2.1. Frequency Plan Tab

1. Configure Input Frequency as 10 MHz using differential inputs IN1/2.
2. Configure Output Frequencies for CLK0 as 10 MHz and CLK3 as 10 MHz. Use the skip button to leave CLK1 and CLK2 blank.
3. Click on “Create Plan”, and “Apply Values to Register Map”. This will download the register setting to the evaluation board.

A screen shot of the resulting Frequency Plan Tab is shown in Figure 3.

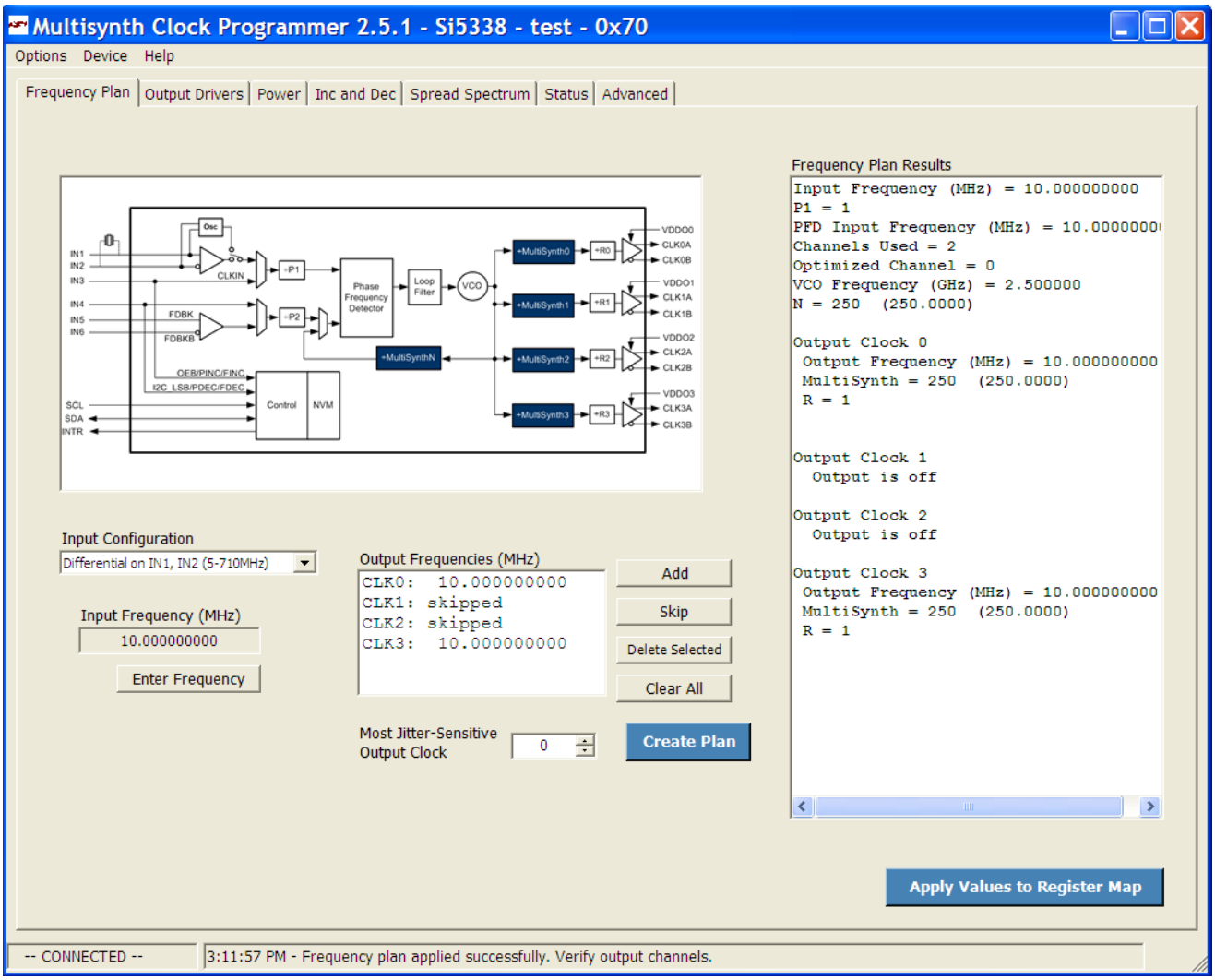


Figure 3. Frequency Plan Tab

2.2. Output Drivers Tab

1. Turn on driver power to CLK0 and CLK3.
2. Set the output type for CLK0 and CLK3. In this example, 3.3 V LVDS was chosen.
3. Set the “Output State when Disabled” on the feedback clock (CLK3) to “Always on”. This is very important—the PLL will not lock otherwise.
4. Click on “Apply Values to Register Map” to download the register setting to the evaluation board.

A screen shot of the resulting Frequency Plan Tab is shown in Figure 4.

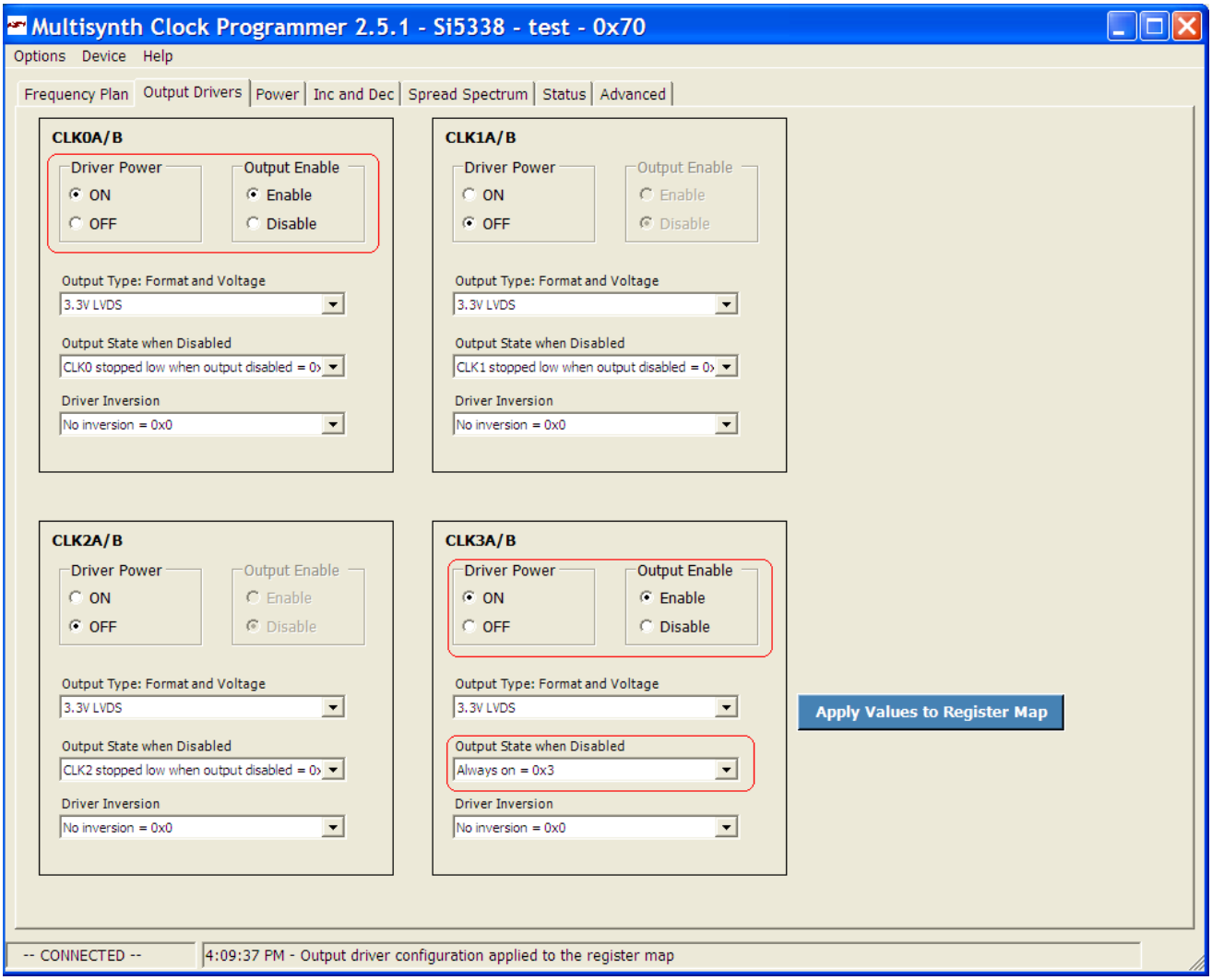


Figure 4. Output Driver Tab

2.3. Advanced Tab

1. Set the feedback path to a differential input (IN5/6).
2. Enter the feedback input frequency as 10 MHz.
3. Select “divfbclk” on input mux.
4. Select “ext” feedback mode.
5. Click on “Apply Values to Register Map.”

A screen shot of the resulting Frequency Plan Tab is shown in Figure 5.

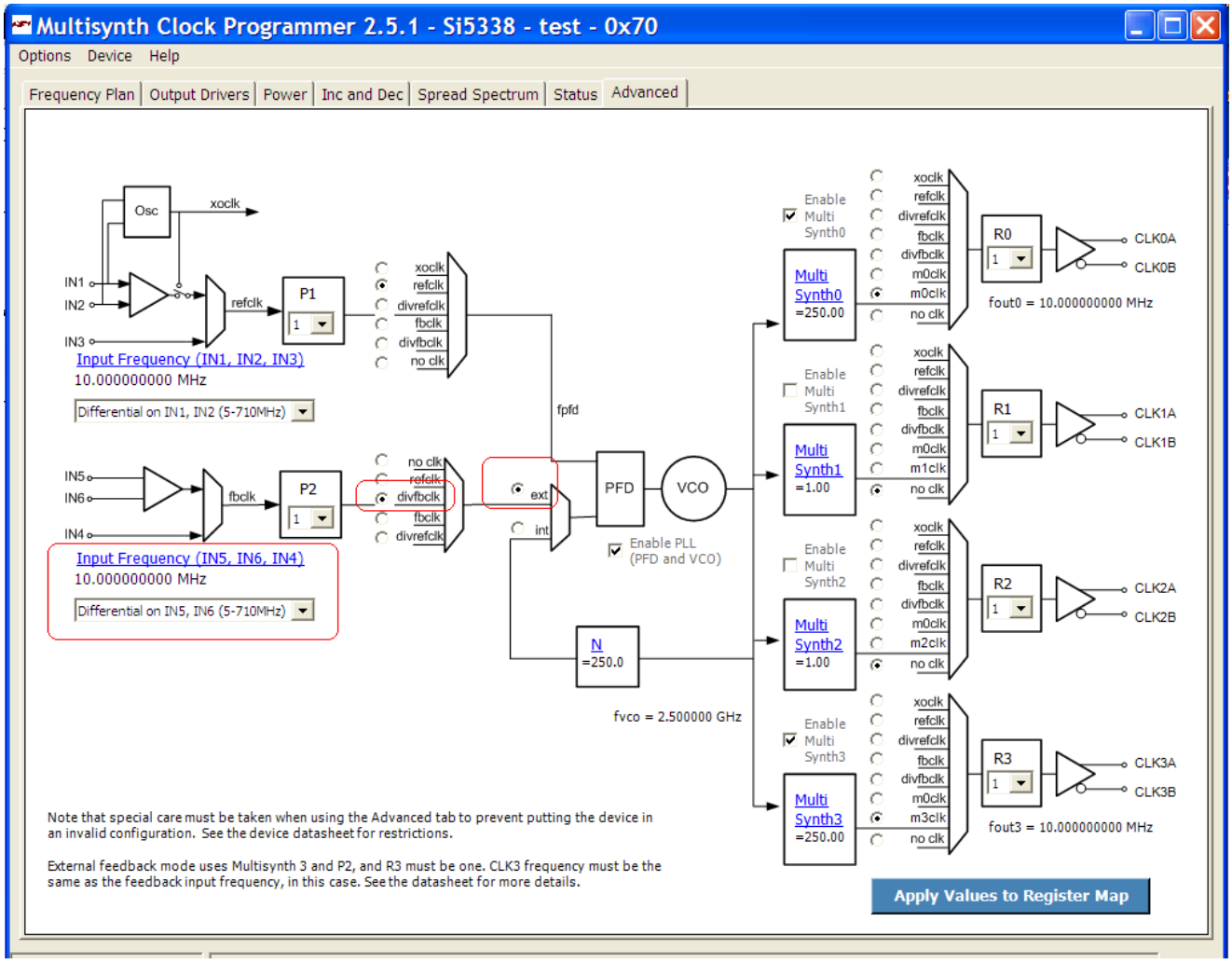


Figure 5. Advanced Tab

3. Conclusion

The Si5338 evaluation board should now generate an output clock with minimal skew with its input clock.

NOTES:

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